Export Controls and Endogenous Upstream Price Incidence:

The Chip War's Consequences for the AI-chip Industry

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The Setting

The event

October 2022: the US restricts NVIDIA's AI chip exports to China.

NVIDIA can no longer sell its flagship accelerators to its largest market.

The input

AI chips require **High-Bandwidth Memory (HBM)** — a specialized component sourced under long-term bilateral contracts from three suppliers: SK hynix, Samsung, and Micron.

The question

When demand from one buyer collapses,

what happens to the prices that suppliers charge — to that buyer, and to its rivals?

Standard trade models give a clear answer. The data say something different.

Roadmap

1. **Motivation**
   Why existing models fail
2. **Stylized Facts**
   Four empirical patterns
3. **Model**
   Bilateral oligopoly + Nash-in-Nash bargaining
4. **Theory**
   Shock-absorber & spillover mechanisms
5. **Calibration**
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6. **Results**
   Price decomposition; non-neutrality
7. **Robustness**
   Bargaining weights & elasticities
8. **Policy Implications**
   Three lessons for concentrated GVCs
9. **Conclusion**
   Summary and open questions

Section 1

Motivation

Why existing models fail

The Standard View of Export Controls

Textbook pass-through identity:

$$d\ln p_d^D = \underbrace{d\ln \tau_d}_{\text{statutory wedge}} + \underbrace{0}_{\substack{\text{upstream price} \\ \text{(competitive} \to \text{fixed)}}} + \underbrace{d\ln \mu_d}_{\text{markup}}$$

Standard assumptions:

- Input market is competitive
- Upstream price = market-clearing price
- Trade shock raises buyer's cost one-for-one
- Rival buyer is unaffected

Implication

An export control on NVIDIA should:

1. Raise NVIDIA's costs
2. Leave AMD's costs unchanged

This logic breaks down when both sides have market power.

What's Special: Bilateral Oligopoly

The AI accelerator supply chain is among the most concentrated bilateral markets on record.[1]

Upstream: HBM suppliers (Q2 2025)

Supplier	Share
SK hynix	62.0%
Samsung	17.0%
Micron	21.0%
HHI	**4,574**

Counterpoint Research Q2 2025. HHI > 2,500 = highly concentrated (DoJ)

Downstream: AI accelerators (2024–25)

Designer	Share
NVIDIA	$\geq 80\%$
AMD	$\sim 20\%$
HHI[†]	**6,400**

[†]Lower bound: NVIDIA ≥80%, AMD ~20%.

In this environment, input prices are **negotiated link by link**, not determined by global supply and demand.

[1]Sources: Counterpoint Research Q2 2025 (upstream); analyst consensus (downstream).

MCP–IC (HS 8542323000) Serves as a Clean HBM Proxy in Korean Customs Data

Identification challenge: HBM has no dedicated long-standing HS code.

Solution: Multi-Chip Package IC (MCP–IC, HS 8542323000)

- Records HBM-attached accelerator modules
- HS10 $\times$ destination $\times$ month granularity

Controls: DRAM and NAND Flash

- Share upstream wafer inputs and macro-demand drivers
- Do *not* face AI-accelerator integration constraints

Active link network (mid-2024)

Downstream	Upstream
NVIDIA	SK hynix
NVIDIA	Micron
AMD	Samsung
AMD	Micron

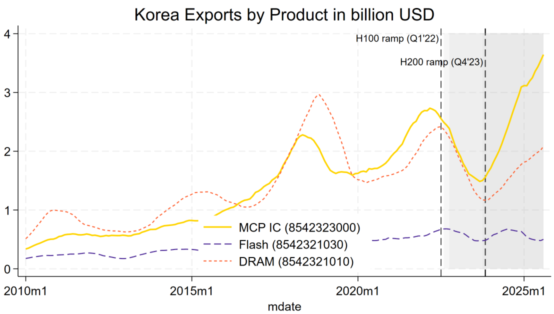
Samsung not yet qualified for NVIDIA — the "non-neutrality" experiment

Section 2

Stylized Facts

Four empirical patterns from Korean customs data

Fact 1: The MCP–IC Boom



Shaded: Notice (light), Oct 7 regime (medium), 2023 update (dark). Tech: HBM3 ramp (thin line).

You might think: "HBM exports follow the memory cycle."

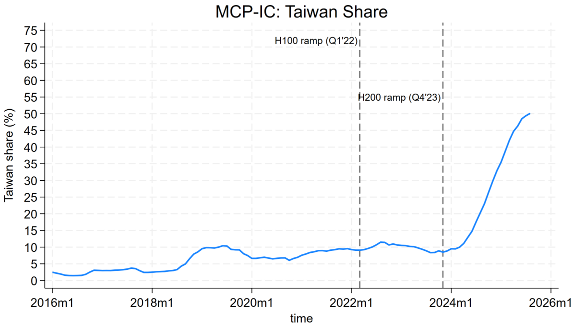
True, but:

- DRAM and NAND Flash share cyclical patterns with MCP–IC
- MCP–IC **accelerates far faster** from 2022 onward — AI demand layered on top of cycle
- By 2025: MCP–IC ≈ **6% of Korea's total exports** (Korea Customs Service)

Takeaway

MCP–IC shares memory-cycle, but its level surge is AI-specific.
→ A valid proxy for HBM demand.

Fact 2: Reallocation Toward Taiwan



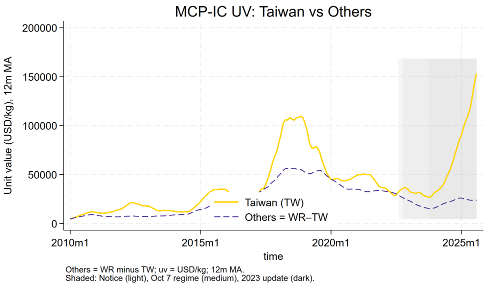
Network rigidity in destination structure:

- Taiwan stable ~15% through 2021; surges post-HBM3/H100 ramp
- China + HK: 29–37% (2020–22) → 9% (2025)
- Taiwan: 7–11% (2020–22) → 52% (2025)

Model translation

Buyers are tethered to production hubs — not costlessly mobile.

Fact 3: Law of One Price Fails



Unit values: Taiwan vs. rest of world

You might think: "HBM prices equalize across buyers via arbitrage."

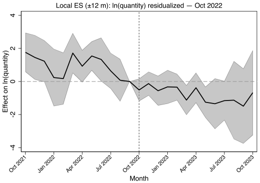
But actually:

- Taiwan-bound MCP–IC: **peak >$200,000/kg**
- Persistent, widening premium vs. other destinations
- Within Taiwan: MCP–IC diverges sharply from DRAM and NAND Flash

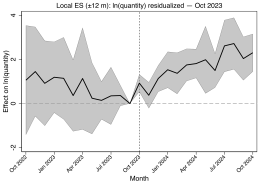
Implication

If competitive, arbitrage would equate prices. Instead, prices are **relationship-specific**; this motivates Nash-in-Nash.

Fact 4: Export Controls Reshaped the Taiwan Corridor



Oct 2022: quantity



Oct 2023: quantity

Taiwan × MCP–IC vs. DRAM/NAND controls. Residualized for trends, seasonality, AI demand.

Heterogeneous τ_d, opposite signs:

- Oct 2022: broad FDP Rule ⇒ heavier effective τ on non-NVIDIA (China-dependent assembly)

  ⇒ aggregate input cost ↑ ⇒ **contraction**
- Oct 2023: A800/H800 closure ⇒ heavier τ on NVIDIA

  Shock-absorber dampens P^{M}_{NV}; AMD rerouted

  ⇒ **expansion**

Different relative τ_d ⇒ different equilibrium response.
Descriptive targets for calibration (pre-trends present).

⇒ *Uniform-tariff models cannot generate this sign reversal.*

▸ B2: ES specification ▸ B3: Pre-trends ▸ B4: Stacked ES

Section 3
Model

Bilateral oligopoly with Nash-in-Nash bargaining

Model: Bargaining Precedes Competition — Three Stages

A static bilateral oligopoly with Nash-in-Nash input bargaining.

Three-stage timeline:

1. **Bargaining.** Each pair (d, u) negotiates p^U_{du} bilaterally. NiN = fixed point.
2. **Cost realization.** Prices pin down unit costs C^D_d.
3. **Cournot competition.** Downstream firms choose quantities.

Design choices:

- **Downstream demand:** CES with curvature σ (firm substitution) and ρ (composite elasticity)
- **Upstream technology:** CES aggregator with elasticity η (input substitution)

$$P^M_d = \left(\sum_u \left(p^U_{du}\right)^{1-\eta} \right)^{\frac{1}{1-\eta}}$$

- **Passive beliefs:** Breakdown of (d, u): all other agreements hold.
- **Export control:** Buyer-specific wedge τ_d in $C^D_d = (\tau_d / z_d) P^M_d$

NiN: Each Link Maximizes Joint Surplus, Taking Others Fixed

For each active pair (d, u), firms maximize the Nash product of incremental surpluses:

$$NP_{du}(p^U_{du}; p^U_{-(d,u)}) = (\Pi^D_d - \Pi^D_{d,-u})^{\gamma_d} \times (\Pi^U_u - \Pi^U_{-d,u})^{1-\gamma_d}$$

Key objects:

- $\Pi^D_d - \Pi^D_{d,-u}$: buyer's incremental surplus (profit gain from this link)
- $\Pi^U_u - \Pi^U_{-d,u}$: seller's incremental surplus (revenue on this link, net of cost)
- $\gamma_d \in (0, 1)$: downstream bargaining weight

Passive beliefs

$\Pi^D_{d,-u}$: payoff if link (d, u) is removed, **all others fixed**

Standard IO assumption (Crawford-Yurukoglu; Collard-Wexler)

NiN equilibrium: price vector $\{p^{U*}_{du}\}$ that simultaneously solves all bilateral problems — the fixed point of link-by-link first-order conditions.

B1: Formal definition

CES Input Bundle: Export Controls Raise Downstream Costs Directly

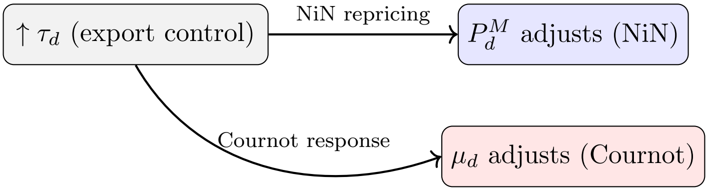
Upstream input bundle: downstream firm d aggregates inputs with CES elasticity η:

$$P_d^M = \left(\sum_u \left(p_{du}^U\right)^{1-\eta}\right)^{1/(1-\eta)}$$

Unit delivery cost:

$$C_d^D = \frac{\tau_d}{z_d} P_d^M$$

Export control enters as an **increase in** τ_d — a buyer-specific cost wedge.



B5: Why CES, not BLP?

Price Decomposition: Four Pass-Through Channels Operate Simultaneously

Total pass-through from $\tau_d \to p_d^D$, where $\ln p_d^D = \ln \mu_d + \ln \tau_d - \ln z_d + \ln P_d^M$:

$$\boxed{d \ln p_d^D = \underbrace{d \ln \tau_d}_{\text{Statutory wedge}} + \underbrace{d \ln P_d^M}_{\text{Upstream price channel (\textbf{new})}} + \underbrace{d \ln \mu_d}_{\text{Markup channel}}}$$

Standard model: $d \ln P_d^M = 0$
All policy effects run through τ_d and μ_d only.

This paper: $d \ln P_d^M \neq 0$
NiN bargaining reprices each link when τ_d changes.

The central question: What is the sign of $d \ln P_d^M / d \ln \tau_d$?

Model: Multinational and Multi-Country Extension

The baseline extends to multi-plant, multi-country settings. Firm d operates plants $i \in \mathcal{I}_d$ across countries; plant i faces iceberg trade costs $\tau_{n'i}$ to serve destination n' (distinct from the policy wedge τ_d).

Plant-level cost:

$$C^{D,(n(i))}_{n',i,d} = \frac{\tau_{n'i}}{z_d} P^M_{i,d}$$

Firm-level profit:

$$\Pi^D_d = \sum_{n'} \sum_{i \in \mathcal{I}_d} \left(p^D_{n',i,d} - C^D_{n',i,d}\right) q^D_{n',i,d}$$

Bargaining is at the plant-pair level (i, j, d, u).

Key result

Cournot sign patterns and diagonal dominance are **unchanged** under multi-country extension. All existence, uniqueness, and comparative-statics results from the baseline carry over.

In the empirical setting: HBM plants in Korea; accelerator assembly in Taiwan; end markets worldwide.

B6: Multi-country extension

Section 4

Theory

Shock-absorber and spillover mechanisms

Theory: Uniqueness of NiN Equilibrium

Proposition 1 (Existence and Uniqueness)

Under demand curvature conditions and a bargaining-weight ceiling $\gamma_d < \bar{\gamma}$, the NiN price vector $\{p_{du}^{U*}\}$ exists and is the unique fixed point of the stacked first-order conditions.

Key ingredient: The Jacobian J of the stacked log-Nash gradient is a **P-matrix**.

P-matrix $\Rightarrow$:

- Unique zero of $g(\mathbf{p}^U) = 0$
- $-J^{-1} \geq 0$ entrywise (M-matrix property)
- Comparative statics are well-defined and sign-coherent

Ceiling condition

$$\bar{\gamma} = 1 - \frac{1}{\eta}\Big(\frac{1}{\rho} - \frac{1}{\sigma}\Big)$$

Calibrated: $\gamma = 0.5 \ll \bar{\gamma} \approx 0.967$.
Proposition applies analytically.

Why it matters: Without uniqueness, comparative statics are meaningless.

B7: P-matrix proof

Theory: The Shock-Absorber Mechanism

You might think: "If NVIDIA is hit, do suppliers raise or lower prices? The direction seems ambiguous."

The sign is pinned by a curvature condition:

Remark (Shock-absorber sign)

$\dfrac{\partial \ln p_{du}^{U*}}{\partial \ln \tau_d} < 0$ (**deflationary**) if and only if $\Delta\Pi_d^D$ is **log-submodular** in (p_{du}^U, τ_d).

Intuition: Squeezing buyer profit makes cheap inputs *disproportionately* valuable — supplier lowers p_{du}^U to protect the relationship.

Cf. Chipty-Snyder (1999) log-submodularity condition for buyer power.

Empirical verdict

Condition holds at baseline calibration ($\sigma = 6$, $\eta = 2.5$, $\gamma = 0.5$).
Shock-absorber sign confirmed at 6/9 grid points in sensitivity table.

Theory: Cross-Buyer Spillover

You might think: "NVIDIA's problem is NVIDIA's problem. AMD is unaffected."

But actually, the M-matrix structure forces a positive cross-effect:

Corollary (Cross-buyer spillover)

If buyers d and d' share at least one active supplier u, then:

$$\frac{\partial \ln p^{U*}_{d'u}}{\partial \ln \tau_d} > 0 \qquad \Rightarrow \qquad \frac{\partial \ln P^{M}_{d'}}{\partial \ln \tau_d} > 0$$

A tighter wedge on d **raises** rival d''s input costs.

Mechanism:
NVIDIA's demand contracts → shared supplier (Micron) has spare capacity → AMD's volume rises → AMD's bargaining position **weakens** → AMD pays more.

Same policy shock. Opposite direction. For two buyers sharing one supplier.

Section 5

Calibration

Anchored to the HBM supply chain

Calibration: Six Parameters Pinned to HBM Market and Trade Data

Symbol	Description	Value
σ	Downstream demand curvature	6.0
ρ	Composite elasticity	4.0
η	Input substitution	2.5
D	# downstream firms	2
U	# upstream firms	3
$\boldsymbol{\gamma}$	Bargaining weights	(0.5, 0.5)
$\boldsymbol{\tau}$	Pre-shock wedges	1 (normalized)
$\boldsymbol{z}_u$	Upstream productivity	calibrated
z_{NV}	NVIDIA productivity	calibrated

Elasticities from literature; $\boldsymbol{z}_u$: Counterpoint Research Q2 2025 HBM shares (LM calibration); z_{NV}: NVIDIA $\geq 80\%$ share.

Goal

Not a perfect structural fit — a **transparent, semi-structured** environment anchored to observed market structure.

Productivities $\boldsymbol{z}_d$, $\boldsymbol{z}_u$: joint LM calibration — z_d targets downstream shares, z_u targets upstream (HBM) shares; equilibrium solver called at each step.

Symmetric γ: deliberately conservative. Isolates network position from bargaining leverage. Heterogeneous sweeps follow.

Calibration: Model Matches Market Shares and Upstream Price Premium

HBM vendor shares: data vs. model

Supplier	Data (%)	Model (%)	
SK hynix	62.0	62.0	✓
Samsung	17.0	17.0	✓
Micron	21.0	21.0	✓

Source: Counterpoint Research Q2 2025.

Samsung: Q2 2025 Counterpoint data (17%) reflects period before Samsung qualified for NVIDIA HBM3e — consistent with AMD-only adjacency in baseline.

Internal consistency

- NiN algorithm: converges <300 iterations, residual $< 10^{-8}$
- Individual rationality: all firms weakly exceed disagreement payoffs
- Ceiling condition: $\gamma = 0.5 \ll \bar{\gamma} \approx 0.967$ $\Rightarrow$ Prop. 1 applies analytically
- NVIDIA/AMD output ratio $\approx$ 4:1 (✓)

$\Rightarrow$ *Calibration passes all internal checks. We now use the calibrated equilibrium to evaluate export-control scenarios.*

Section 6

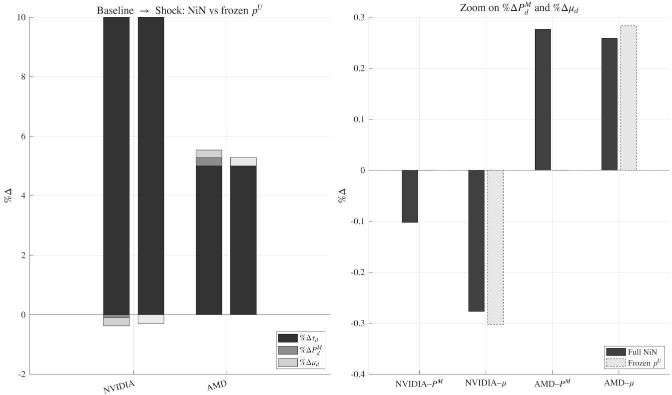
Results

Price decomposition and network non-neutrality

Results: Price Decomposition — Scenario A

Scenario A: NVIDIA +10% wedge, AMD +5% wedge ($\Delta\tau_{\text{NV}} > \Delta\tau_{\text{AMD}}$)

($\%\Delta X \equiv 100 \cdot d\ln X$)



Key: NVIDIA input price ↓ (shock absorber) | AMD input price ↑ (cross-buyer spillover)

Dark bars = full NiN equilibrium. Light bars = frozen-input benchmark (competitive inputs).

The **gap** between dark and light bars isolates the endogenous upstream repricing channel.

Results: Scenario A — Reading the Numbers

Endogenous channels (full NiN):

- $\%\Delta P^M_{\text{NV}} \approx -0.16\%$ $\downarrow$ shock absorber
- $\%\Delta P^M_{\text{AMD}} \approx +0.22\%$ $\uparrow$ cross-buyer spillover
- $\%\Delta \mu_{\text{NV}} \approx -0.82\%$
- $\%\Delta \mu_{\text{AMD}} \approx +0.22\%$

Ratio: Input-price channel $\approx 1/5$ of markup response for NVIDIA.

What the frozen benchmark misses

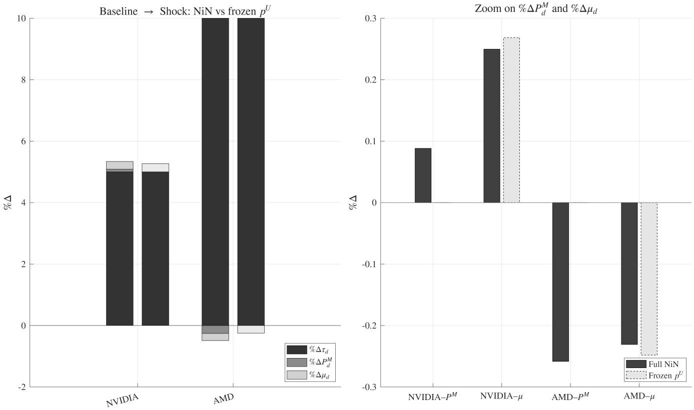
Setting $d \ln P^M_d = 0$ (competitive input market):

- Misses NVIDIA's deflationary relief
- Misses AMD's inflationary spillover
- Overstates competitive reallocation from NVIDIA to AMD

Standard models overestimate the competitive reallocation caused by export controls.

Results: Price Decomposition — Scenario B (Sign Reversal)

Scenario B: NVIDIA +5% wedge, AMD +10% wedge ($\Delta\tau_{\text{AMD}} > \Delta\tau_{\text{NV}}$)



Key: AMD input price ↓ (shock absorber) | NVIDIA input price ↑ (cross-buyer spillover)

Same model. Same parameters. Same network.

Flip the exposed buyer → flip the sign pattern.

Results: Scenario B — Reading the Numbers

Endogenous channels (full NiN):

- $\%\Delta P^M_{\text{AMD}} \approx -0.09\%$ ↓ shock absorber
- $\%\Delta P^M_{\text{NV}} \approx +0.09\%$ ↑ cross-buyer spillover
- $\%\Delta \mu_{\text{AMD}} \approx -0.24\%$
- $\%\Delta \mu_{\text{NV}} \approx +0.13\%$

Same mechanism, reversed polarity.
AMD is smaller — magnitudes attenuated relative to Scenario A.

What the frozen benchmark misses

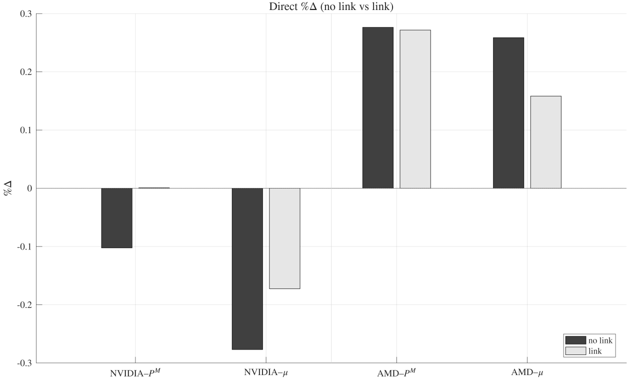
Setting $d \ln P^M_d = 0$ (competitive):

- Misses AMD's deflationary relief
- Misses NVIDIA's inflationary spillover
- Overstates AMD's competitive advantage

The shock-absorber is structural — it follows the exposed buyer, regardless of identity.

Results: Network Non-Neutrality

You might think: "Adding one more HBM supplier for NVIDIA (Samsung) is a modest improvement."



Dark bars = baseline (no Samsung–NVIDIA link). Light bars = Samsung–NVIDIA link active.

Scenario A (NVIDIA hit harder):

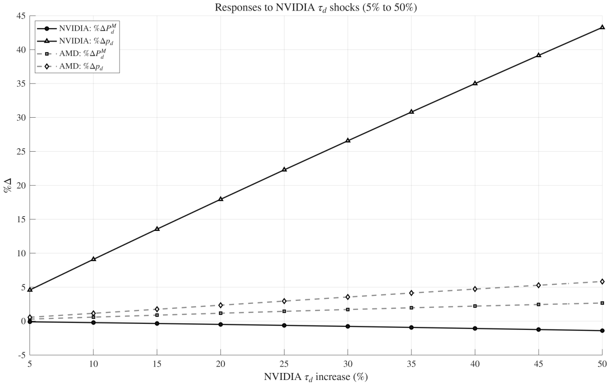
- NVIDIA markup increase dampened ∼**46%**
- P^M_{NV} adjustment nearly **disappears**
- AMD's spillover also attenuated

Policy insight

Supply-chain diversification **fundamentally reshapes incidence** — not a marginal improvement.

B8: Scenario B recalibration

Results: NVIDIA Wedge Sweep



Key patterns:

- NVIDIA's price pass-through ≈ 1 (deflationary offset from NiN)
- AMD's input costs rise **monotonically** as τ_{NV} widens
- Spillover is **not a corner case** — accumulates with shock magnitude

Interpretation

Even modest NVIDIA controls raise AMD's HBM costs. The channels grow together as the wedge widens.

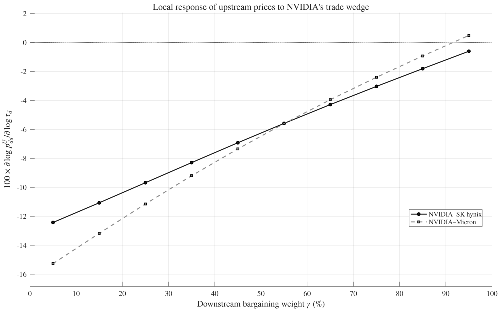
B9: Elasticity grid

Section 7

Robustness

Bargaining weights and structural elasticities

Shock-Absorber Sign Holds for $\gamma < 0.85$ — Comfortably Met



Local derivative $\partial \ln p^{U}_{\mathrm{NV},u} / \partial \ln \tau_{\mathrm{NV}}$ vs. γ_{NV}

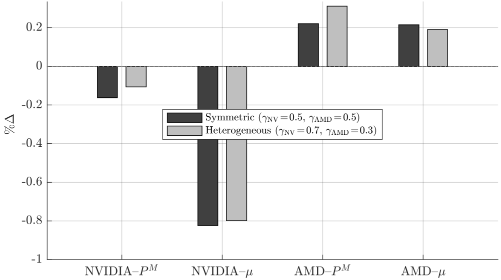
You might think: "Symmetric $\gamma = 0.5$ is unrealistic — NVIDIA must have more bargaining power."

- Sign **stable**: $\gamma_{\mathrm{NV}} \in (0, \approx 0.85)$
- **Reverses** above $\gamma \approx 0.85$: very high buyer power $\Rightarrow$ controls *amplify* rather than dampen
- Magnitude scales smoothly — no knife-edge

Robustness:
$\gamma \in \{0.30,\ 0.50,\ 0.70\}$

Direction preserved for all three values.
NVIDIA's $\geq$80% share puts empirical γ well below the 0.85 threshold.

Heterogeneous γ: Results Are Not Driven by Symmetric Bargaining



Price decomposition: symmetric vs. heterogeneous γ (Scenario A: $\tau_{\text{NV}} + 10\%$, $\tau_{\text{AMD}} + 5\%$)

Dark: symmetric ($\gamma_{\text{NV}} = \gamma_{\text{AMD}} = 0.5$). Light: heterogeneous ($\gamma_{\text{NV}} = 0.7$, $\gamma_{\text{AMD}} = 0.3$).

With $\gamma_{\text{NV}} = 0.7$, $\gamma_{\text{AMD}} = 0.3$ (re-calibrated productivities):

- $\%\Delta P^M_{\text{NV}}$: $-0.163\% \to -0.107\%$ (slightly less deflationary)
- $\%\Delta P^M_{\text{AMD}}$: $+0.221\% \to +0.311\%$ (spillover amplified)
- Markup responses nearly **identical** across configurations

Verdict

Qualitative sign pattern (*NVIDIA absorbs, AMD spills over*) is **preserved** under plausible heterogeneity in bilateral bargaining leverage.

Section 8

Policy Implications

Three lessons for concentrated GVCs

Three Lessons: Controls Self-Attenuate in Concentrated Supply Chains

Three lessons for export controls in concentrated supply chains:

1. **Standard models overestimate control efficacy.**
   Shock-absorber offsets $\approx 1/5$ of markup response. Requires log-submodularity of buyer surplus **and** $\gamma < 0.85$ — both hold at baseline.

2. **Monitoring requires upstream input prices, not just downstream output quantities.**
   Repricing runs through **confidential bilateral contracts** — invisible in aggregate customs data.

3. **Controls in concentrated supply chains self-attenuate.**
   Lower effective costs reduce lobbying pressure and erode the policy's strategic bite simultaneously.

Policy Implications: Supply-Chain Diversification

The network non-neutrality result has a direct policy reading

Qualifying Samsung for NVIDIA is not just a backup — it **reshapes equilibrium incidence** of any future control shock.

With Samsung–NVIDIA link active:

- NVIDIA's markup increase dampened $\sim$46%
- NVIDIA's input-price relief increases further
- AMD's spillover cost reduced

Implication for industrial policy:

- Resilience programs (e.g., CHIPS Act) are not merely about redundancy
- They alter the *bargaining geometry* — redistributing the burden of future shocks

In concentrated GVCs, incidence depends on the configuration of active contracts — not just the wedge.

Section 9

Conclusion

What we found and why it matters

Conclusion: Endogenous Upstream Pricing Reshapes Incidence of Trade Policy

The paper showed:

- **Deflationary channel:** log-submodularity of buyer surplus in (p^U, τ) — with a boundary: sign reverses at $\gamma \gtrsim 0.85$
- **AMD spillover:** M-matrix structure forces positive cross-effects on shared links
- **Network non-neutrality:** incidence depends on the adjacency matrix, not just the wedge
- **Quantitatively:** upstream channel offsets ∼1/5 of markup response; standard models overestimate reallocation

The general lesson

In concentrated GVCs, buyer-specific trade policy is blunted if one ignores the input market. Must take seriously:

1. Vertical structure (who is linked to whom)
2. Bargaining power (γ)
3. Endogenous repricing of bottleneck inputs (P_d^M)

Open questions: Dynamic investment and entry. Does the shock-absorber weaken as buyers diversify and more upstream suppliers qualify for top-tier links?

Backup Slides

B1: Bilateral Oligopoly Equilibrium (Formal Definition)

A collection $\{\mathbf{p}^{U*}, \mathbf{C}^D, \mathbf{q}^{D*}\}$ is a **bilateral oligopoly equilibrium** if:

1. **NiN bargaining.** The input-price vector $\mathbf{p}^{U*}$ solves the NiN Nash product for every active pair (d, u), with passive beliefs: upon breakdown of (d, u), all other pairs maintain their agreements.
2. **Unit-cost mapping.** Given $\mathbf{p}^{U*}$:

$$C_d^D = \frac{\tau_d}{z_d} P_d^M(\mathbf{p}^{U*}), \qquad P_d^M = \left(\sum_u (p_{du}^{U*})^{1-\eta}\right)^{1/(1-\eta)}$$

3. **Cournot competition.** Each downstream firm chooses q_d^D to maximize $(p_d^D - C_d^D)q_d^D$, taking rivals' quantities as given. Equilibrium $\mathbf{q}^{D*}$ is unique under regularity conditions.

Existence and uniqueness: Follows from the P-matrix structure of the stacked NiN Jacobian (Proposition 1) and Rosen's theorem.

Back

B2: Event Study — Specification Details

Design:

- **Treated unit:** Taiwan × MCP–IC (HS 8542323000)
- **Controls:** DRAM (HS 8542321010), NAND (HS 8542321030) — same upstream wafer inputs, different vertical
- **Window:** ±12 months around each event

Fixed effects & residualization:

- Destination × month FE
- Item × month FE
- Unit trend + unit × month seasonality
- AI demand control: demeaned log NVIDIA data-center revenue × unit exposure

What the ES does *not* do:

- Does *not* identify bilateral price divergence ($\downarrow p^U_{\mathrm{NV}}, \uparrow p^U_{\mathrm{AMD}}$) — these are mixed in aggregate unit values
- Does *not* deliver causal estimates — pre-trends present

What the ES *does* do:

- Documents *direction and magnitude* of aggregate trade flow disruption
- Establishes the opposite-signed pattern across regimes
- Provides calibration targets for the structural model

Back

B3: Event Study — Pre-Trend Discussion

Why pre-trends are non-zero (and why this is not a problem):

- Oct 2022 event: $F(11, \cdot) = 4.63$ ($p < 0.001$) for export value. Pre-trends reflect **real economic dynamics** — SK hynix's HBM3 development ramp differentially affected MCP–IC before any policy.
- The immediate pre-period ($k = -7$ to $k = -2$): $F(6, \cdot) = 1.25$ ($p = 0.294$) for value — relatively flat before the policy window.
- Oct 2023 event: quantity pre-trend reflects **anticipatory front-loading** ahead of announced deadline.

Identification strategy

The event-study is read as a **descriptive dynamic profile** of the Taiwan $\times$ MCP–IC cell through each policy window — not as a parallel-trends causal estimate. Causal identification comes from the structural model's equilibrium moment conditions.

Robustness: stacked event studies (Appendix A.2) and restricted samples (Appendix A.3) confirm results are not driven by residualization choices.

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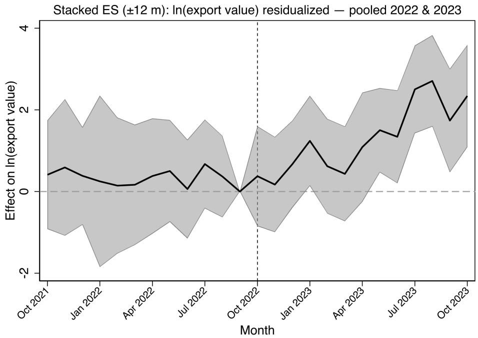
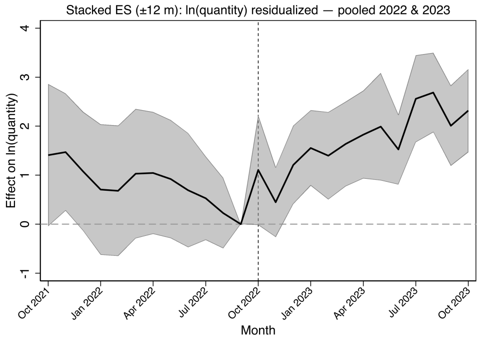
B4: Stacked Event Study and Alternative Designs



Stacked: ln(export value)



Stacked: ln(quantity)

Stacked design pools both events with a common pre-period window. Confirms that the opposite-signed responses across regimes are not an artifact of the local event-study design.

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B5: Why CES, Not BLP?

BLP is designed for consumer markets:

- Many buyers with heterogeneous tastes
- Variation across many markets to identify taste shocks
- Market power from product differentiation on the demand side

None of these hold here.

Three reasons CES fits:

1. **Intermediate goods.** NVIDIA and AMD source HBM to engineering specs — substitution is across technically similar suppliers, not differentiated consumer varieties.
2. **Market power is bilateral.** The key mechanism is NiN bargaining between two concentrated sides. Adding BLP curvature on the demand side would not generate the cross-buyer price linkage — it is a supply-side phenomenon.
3. **Closed form.** CES delivers the price index P_d^M in closed form, which is essential for the analytical price decomposition. BLP does not integrate

B6: Multi-Country / Multinational Extension

Setup: Firm d operates plants $i \in \mathcal{I}_d$ across countries; firm u operates plants $j \in \mathcal{J}_u$. Bargaining at plant-pair level (i, j, d, u).

Cost structure:

$$P^M_{i,d} = \left(\sum_u \sum_{j \in \mathcal{J}_u} (p^U_{ij,du})^{1-\eta} \right)^{1/(1-\eta)}, \qquad C^D_{n',i,d} = \frac{\tau_{n'i}}{z_d} P^M_{i,d}$$

Preservation of results:

- Plant-level CES preserves nested-CES curvature bounds (Lemma A1 applies plant-by-plant)
- Cournot block remains unique and diagonally dominant
- Stacked plant-pair Jacobian J^{MC} remains a P-matrix
- Comparative statics sign pattern is unchanged

Empirical mapping:

- HBM plants: Korea (SK hynix, Samsung, Micron Korea fabs)
- Accelerator assembly: Taiwan (TSMC CoWoS)
- End markets: global (incl. China, pre-control)

Export control = increase in $\tau_{n'i}$ for China-facing plants of NVIDIA.

B7: P-Matrix Proof Sketch

Proposition 1 proof sketch (three steps):

Step 1 (Cournot uniqueness): Nested-CES demand curvature and condition $\sigma \leq 2\rho$ imply the Cournot block is a game of strategic substitutes with diagonally dominant best replies. Unique Cournot equilibrium for any fixed $\mathbf{p}^U$.

Step 2 (One-link concavity): Fixing all other prices, buyer's incremental surplus $\Delta\Pi_d^D$ is strictly decreasing and log-concave in own price (under bargaining-weight ceiling). Seller's incremental surplus is strictly log-concave. $\Rightarrow$ Unique one-link NiN solution.

Step 3 (Diagonal dominance $\Rightarrow$ P-matrix): Stack all link FOCs. Own-link curvature admits uniform lower bound:

$$-J_{LL} \geq \frac{\gamma_d \underline{\kappa}_d^D + (1-\gamma_d)\underline{\kappa}_d^U}{p_L^2}$$

Cross-link effects (via CES index P_d^M and Cournot) are uniformly bounded. Gershgorin argument on symmetrized $S = (RJ + J^\top R)/2$ gives $S \prec 0$, hence J is a P-matrix. Rosen's theorem delivers unique fixed point $\mathbf{p}^{U*}$.

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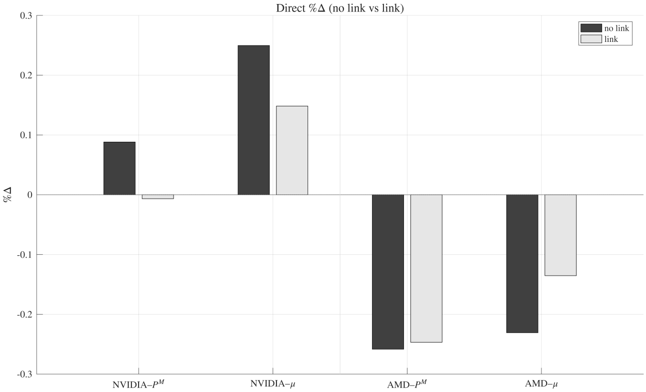
B8: Non-Neutrality — Recalibrated (Scenario B)



Dark bars = baseline (no Samsung–NVIDIA link). Light bars = Samsung–NVIDIA link active.

Scenario B (AMD hit harder): Adding Samsung–NVIDIA link **amplifies** AMD's input price and markup effects while **muting** NVIDIA's. The same extensive-margin change has opposite implications depending on which buyer is most constrained.

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B9: Elasticity Sensitivity Grid

Key finding: Upstream repricing channel essentially inactive at $\sigma = 4$; activates at $\sigma = 6$ and strengthens at $\sigma = 8$.

		NVIDIA		AMD	
σ	η	$\%\Delta P^M$	$\%\Delta\mu$	$\%\Delta P^M$	$\%\Delta\mu$
4	1.5	≈ 0	-0.47	≈ 0	-0.12
4	2.5	≈ 0	-0.47	≈ 0	-0.12
4	4.0	≈ 0	-0.47	≈ 0	-0.12
6	1.5	-0.13	-0.75	$+0.34$	$+0.14$
6	2.5^*	$\mathbf{-0.16}$	$\mathbf{-0.82}$	$\mathbf{+0.22}$	$\mathbf{+0.22}$
6	4.0	-0.14	-0.83	$+0.20$	$+0.22$
8	1.5	-0.08	-0.86	$+0.53$	$+0.22$
8	2.5	-0.27	-1.05	$+0.34$	$+0.42$
8	4.0	-0.25	-1.06	$+0.34$	$+0.43$

Scenario A ($\tau_{\text{NV}} + 10\%$, $\tau_{\text{AMD}} + 5\%$). $\rho = 4.0$ held fixed. Productivities re-calibrated at each grid point.

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